

Ragionato per 14s

TSIA2 Math Practice Test 7

PART A — QUESTIONS

TSIA2-M7-001

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A biology lab uses 3.6 milliliters of reagent to prepare 12 samples. At the same rate, how many milliliters of reagent are needed to prepare 35 samples?

- A) 9.8 milliliters
- B) 10.5 milliliters
- C) 11.2 milliliters
- D) 12.6 milliliters

TSIA2-M7-002

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve for x: $4(x - 6) + 3x = 25$

- A) 5
- B) 6
- C) 7
- D) 9

TSIA2-M7-003

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular bulletin board is 4 feet wide and 6 feet tall. A student wants to cover exactly $\frac{3}{4}$ of the board with posters. How many square feet will be covered?

- A) 12 square feet
- B) 18 square feet
- C) 24 square feet
- D) 32 square feet

TSIA2-M7-004

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The numbers of text messages sent by a student over five days were 14, 22, 19, 31, and 24. What is the median number of text messages sent?

- A) 19
- B) 22

- C) 24
- D) 31

TSIA2-M7-005

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A commuter train ticket costs \$4.80 for a 12-mile trip. What is the cost per mile?

- A) \$0.30
- B) \$0.40
- C) \$0.48
- D) \$0.60

TSIA2-M7-006

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $5(2p - 1) - 3(p + 4)$?

- A) $7p - 17$
- B) $7p + 7$
- C) $13p - 17$
- D) $13p + 7$

TSIA2-M7-007

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A circular rug has a radius of 4 feet. What is the circumference of the rug?

- A) 4π feet
- B) 8π feet
- C) 16π feet
- D) 64π feet

TSIA2-M7-008

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A container has 8 plain muffins, 5 blueberry muffins, and 7 banana muffins. One muffin is selected at random. What is the probability that the muffin is blueberry?

- A) $1/4$
- B) $1/3$
- C) $2/5$
- D) $5/8$

TSIA2-M7-009

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A community center had 240 members last year. This year, it has 276 members. What was the percent increase in membership?

- A) 12%
- B) 15%
- C) 18%
- D) 36%

TSIA2-M7-010

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A plumber charges a fixed service fee of \$45 plus \$38 per hour. If h represents the number of hours and T represents the total charge in dollars, which equation models this situation?

- A) $T = 45h + 38$
- B) $T = 38h + 45$
- C) $T = 83h$
- D) $T = 38(h + 45)$

TSIA2-M7-011

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A cylinder has a radius of 5 centimeters and a height of 8 centimeters. What is the volume of the cylinder?

- A) 40π cubic centimeters
- B) 80π cubic centimeters
- C) 160π cubic centimeters
- D) 200π cubic centimeters

TSIA2-M7-012

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The table shows the number of minutes four students spent on a practice quiz.

Student A: 18

Student B: 26

Student C: 22

Student D: 30

What is the mean time spent on the practice quiz?

- A) 22 minutes
- B) 23 minutes
- C) 24 minutes
- D) 25 minutes

TSIA2-M7-013

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A fundraiser collected \$684 from 36 tickets sold. If each ticket had the same price, what was the price of one ticket?

A) \$17

B) \$18

C) \$19

D) \$21

TSIA2-M7-014

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

For the function $f(x) = 3x^2 - 2x - 4$, what is $f(-2)$?

A) 4

B) 8

C) 12

D) 16

TSIA2-M7-015

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A right triangle has legs of length 10 units and 24 units. What is the length of the hypotenuse?

A) 26 units

B) 28 units

C) 30 units

D) 34 units

TSIA2-M7-016

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A deck of 20 cards has 6 red cards, 9 blue cards, and 5 green cards. Two cards are selected at random without replacement. What is the probability that both cards are green?

A) $\frac{1}{19}$

B) $\frac{1}{16}$

C) $\frac{1}{10}$

D) $\frac{1}{4}$

TSIA2-M7-017

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A recipe calls for $1\frac{1}{4}$ cups of rice to make 5 servings. How many cups of rice are needed to make 14 servings at the same rate?

- A) $2\frac{1}{2}$ cups
- B) 3 cups
- C) $3\frac{1}{2}$ cups
- D) 4 cups

TSIA2-M7-018

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $x^2 - x - 30$?

- A) $(x - 5)(x + 6)$
- B) $(x + 5)(x - 6)$
- C) $(x - 10)(x + 3)$
- D) $(x + 10)(x - 3)$

TSIA2-M7-019

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A square tile has an area of 81 square centimeters. What is the length of one side of the tile?

- A) 8 centimeters
- B) 9 centimeters
- C) 18 centimeters
- D) 40.5 centimeters

TSIA2-M7-020

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A data set has the following values:

6, 8, 10, 10, 12, 14, 18

Which statement is true?

- A) The mean is 10.
- B) The median is 10.
- C) The range is 10.
- D) The mode is 12.

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M7-001 — Correct: B — Correct Answer: 10.5 milliliters — Explanation: The rate is $3.6 \div 12 = 0.3$ milliliter per sample. For 35 samples, $35 \times 0.3 = 10.5$ milliliters.

TSIA2-M7-002 — Correct: C — Correct Answer: 7 — Explanation: Distribute and combine like terms: $4x - 24 + 3x = 25$, so $7x - 24 = 25$. Then $7x = 49$, so $x = 7$.

TSIA2-M7-003 — Correct: B — Correct Answer: 18 square feet — Explanation: The board area is

$4 \times 6 = 24$ square feet. Three-fourths of 24 is 18 square feet.

TSIA2-M7-004 — Correct: B — Correct Answer: 22 — Explanation: In order, the data are 14, 19, 22, 24, and 31. The middle value is 22, so the median is 22.

TSIA2-M7-005 — Correct: B — Correct Answer: \$0.40 — Explanation: Divide the total cost by the number of miles: $\$4.80 \div 12 = \0.40 per mile.

TSIA2-M7-006 — Correct: A — Correct Answer: $7p - 17$ — Explanation: Distribute and combine like terms: $10p - 5 - 3p - 12 = 7p - 17$.

TSIA2-M7-007 — Correct: B — Correct Answer: 8π feet — Explanation: The circumference of a circle is $C = 2\pi r$. With $r = 4$, $C = 8\pi$ feet.

TSIA2-M7-008 — Correct: A — Correct Answer: $1/4$ — Explanation: There are $8 + 5 + 7 = 20$ muffins total, and 5 are blueberry. The probability is $5/20 = 1/4$.

TSIA2-M7-009 — Correct: B — Correct Answer: 15% — Explanation: The increase is $276 - 240 = 36$. The percent increase is $36/240 = 0.15$, or 15%.

TSIA2-M7-010 — Correct: B — Correct Answer: $T = 38h + 45$ — Explanation: The total charge is the hourly charge times the number of hours plus the fixed service fee, so $T = 38h + 45$.

TSIA2-M7-011 — Correct: D — Correct Answer: 200π cubic centimeters — Explanation: The volume of a cylinder is $V = \pi r^2 h$. With $r = 5$ and $h = 8$, $V = \pi(5^2)(8) = 200\pi$ cubic centimeters.

TSIA2-M7-012 — Correct: C — Correct Answer: 24 minutes — Explanation: The total time is $18 + 26 + 22 + 30 = 96$ minutes. Dividing by 4 gives a mean of 24 minutes.

TSIA2-M7-013 — Correct: C — Correct Answer: \$19 — Explanation: Divide the total amount collected by the number of tickets: $\$684 \div 36 = \19 .

TSIA2-M7-014 — Correct: C — Correct Answer: 12 — Explanation: Substitute -2 for x : $f(-2) = 3(-2)^2 - 2(-2) - 4 = 12 + 4 - 4 = 12$.

TSIA2-M7-015 — Correct: A — Correct Answer: 26 units — Explanation: The hypotenuse is $\sqrt{(10^2 + 24^2)} = \sqrt{(100 + 576)} = \sqrt{676} = 26$ units.

TSIA2-M7-016 — Correct: A — Correct Answer: $1/19$ — Explanation: The probability of selecting a green card first is $5/20$. Without replacement, the probability of selecting a green card second is $4/19$. The product is $5/20 \times 4/19 = 1/19$.

TSIA2-M7-017 — Correct: C — Correct Answer: $3 \frac{1}{2}$ cups — Explanation: The recipe uses $1.25 \div 5 = 0.25$ cup per serving. For 14 servings, $14 \times 0.25 = 3.5$ cups, or $3 \frac{1}{2}$ cups.

TSIA2-M7-018 — Correct: B — Correct Answer: $(x + 5)(x - 6)$ — Explanation: The numbers 5 and -6 multiply to -30 and add to -1 , so $x^2 - x - 30 = (x + 5)(x - 6)$.

TSIA2-M7-019 — Correct: B — Correct Answer: 9 centimeters — Explanation: The side length of a square is the square root of its area. $\sqrt{81} = 9$, so the side length is 9 centimeters.

TSIA2-M7-020 — Correct: B — Correct Answer: The median is 10. — Explanation: The data are already in order. With 7 values, the median is the 4th value, which is 10.